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Studierichting: Informatica
Oefeningenreeks 4A nr. 44.2

$$\int \frac{3x^2+x+4}{x^3+x^2+x+1}dx$$

$$x^3+x^2+x+1 = x^2\left(x+1\right)+1\left(x+1\right) = \left(x+1\right)\left(x^2+1\right)$$

$$\frac{3x^2+x+4}{\left(x+1\right)\left(x^2+1\right)} = \frac{A}{x+1}+\frac{Bx+C}{x^2+1}$$

$$A = \left.\frac{3x^2+x+4}{x^2+1}\right|_{x=-1} = 3$$

$$Bi+C = \left.\frac{3x^2+x+4}{x+1}\right|_{x=i} = \frac{3i^2+i+4}{i+1} = \frac{1+i}{1+i} = 1$$

$$\Rightarrow B=0, C=1$$

$$\Rightarrow \int\left(\frac{3}{x+1}+\frac{1}{x^2+1}\right)dx$$

$$= 3\int\frac{dx}{x+1}+\int\frac{dx}{x^2+1}$$

$$= 3\ln\left|x+1\right| + \operatorname{Bgtan} x + k$$

References